

Futureproof

Equity Preservation Mortgage®

Modelling v14d (Optimized)

Model Assumptions & Parameter Risk

Independent Review and Stress Testing

May 2026

Based on the Equity Preservation Mortgage (EPM) v14d (Optimized) Model GBM
(with Stochastic Drift) + Mean Reversion Equity Model (Shevchenko 2026)
50,000-path Monte Carlo Simulation | May 2026

Purpose: This paper sets out the assumptions used in the EPM model, reviews parameters settings and applies stress tests to determine the sensitivity of the model to key parameters. This review should be read in conjunction with *the Actuarial Review Paper* (May 2026).

An empirical review of the Monte-Carlo calibration choices underpinning the EPM v14d Optimized pricing model. This paper quantifies parameter uncertainty on the six material inputs, maps each to headline Probability-of-Deficit (PoD), and combines them into plausible central and adverse scenarios.

Item	Detail
Prepared by	Futureproof internal actuarial — Opus 4.7 review
Date	May 2026
Model version	EPM v14d Model_Optimized (50,000-path Monte Carlo)
Equity model	GBM + mean reversion (Shevchenko 2026)
Cash-rate model	Vasicek (OU) with exact discretization
Scope	Calibration & parameter risk; point-in-time 30-yr contract
Status	Final - Discussion document

Contents

1. Executive Summary
2. Parameter Inventory & Materiality
3. Mean-reversion strength
4. Long-run Cash Rate
5. Collar Cost
6. Equity Drift
7. Wholesale Funding Margin
8. Low-leverage Parameters
9. Combined Parameter Uncertainty
10. Comparison to wholesale investment universe
11. Recommendations
12. Caveats & imitations

1. Executive Summary

The EPM is a **strongly positive-expectation product**. At its base-case, data-calibrated parameters, **92% of 50,000 simulated 30-year paths fully self-fund** — the mortgage offset account grows past the mortgage loan balance, with a typical (median) surplus near **\$1M** shared between Futureproof and the funder. The homeowner draws a guaranteed income of **2% of home value p.a.** throughout the Annuity Period, retains their home, and preserves all home wealth (original equity + property appreciation during loan term) in every outcome, bearing none of the investment risk. The 8.37% minority tail is the complement of the self-funding figure — and it is **fully absorbed by the two-layer LMI + reinsurance structure**, with no deficit going uninsured. The disciplined task of this paper is to test how durable that base-case is to plausible parameter alternatives. The stress scenarios that follow are deliberately conservative (they shift several soft inputs to adverse values at once), and the central finding is that **the layered insurance architecture continues to absorb the tail in every plausible scenario tested** — sized to hold in conditions materially worse than the data-calibrated model base-case.*

The EPM v14d Optimized Model produces a headline per-mortgage Probability of Deficit (PoD) of **8.37% at maturity** (50,000-path Monte Carlo, Pavel's authoritative workbook). PoD throughout this paper refers to this per-mortgage balance-sheet metric at maturity; the insurance cascade (LMI PoC, LMI reinsurance PoC, Portfolio Reinsurance PoC) is detailed in the companion *Model Review* paper. This number is internally consistent with its inputs. The question this paper asks is whether those inputs themselves are robustly supported by the available data.

Our finding is that three of the seventeen model inputs — equity drift μ , mean-reversion strength γ , and the annual collar cost — carry enough empirical uncertainty that plausible alternative calibrations move the headline PoD materially. That is normal for a 30-year equity-linked product and is precisely why the base-case should be paired with conservative stress scenarios for capital sizing — which is what this paper provides.

1.1 The three primary uncertainty drivers

The three primary drives are:

- **Mean-reversion strength γ .** The model uses $\gamma = 0.163$, sourced from Shevchenko (2026). An independent MLE on S&P 500 TR 1988–2024 gives $\gamma_{\text{hat}} = 0.173$ — apparent confirmation. However, the 95% bootstrap confidence interval is $[0.019, 0.447]$, and a likelihood-ratio test against $\gamma = 0$ yields $p = 0.092$, not significant at the 5% level. A small-sample bias study shows that MLE on 36-year samples over-estimates γ by ~ 0.11 on average; the true γ most consistent with the data may sit near 0.05–0.10. PoD is 8.4% at $\gamma = 0.163$, 11.3% at $\gamma = 0.10$, and 38.4% at $\gamma = 0$.
- **Collar cost 0.046%.** The model uses 0.046% annual drag for an 80/140 collar. Black-Scholes pricing at current AU rates, model θ , and 2% dividend yield gives a range of 0.16%–0.44% — 3–10× the model input, before skew and transaction costs. At a central BS estimate of 0.33% annual drag, PoD is roughly 11%; at 1.0%, 25.1%.
- **Equity drift μ 9.2%.** In-sample MLE on 1988–2024 S&P 500 TR gives $\mu_{\text{hat}} = 11.0\%$ and the full-sample arithmetic mean is 12.3%. Both support the model input as historically grounded. Forward-looking estimates that account for current valuation multiples (CAPE-implied forward nominal 5–6%), AU superannuation actuarial assumptions (diversified equities 7–8%) sit materially lower. BlackRock's forward-looking 30 year view of S&P 500 TR is higher at 10% (cf. most recent 30 year index performance of 10.8%) this is reduced by the application of BlackRock's volatility control layer, to 9.2%. (the model base-case). Importantly, based on its portfolio construction of index reference assets. PoD is 8.4% at $\mu = 9.2\%$, 32.2% at $\mu = 8.0\%$, and 60.4% at $\mu = 7.0\%$.

* Combined-scenario PoD: 40% (realistic-central) and 69% (adverse-plausible); five soft inputs shifted jointly — a conservative joint Futureproof | Model Assumptions & Parameter Risk | Internal | May 2026 Page 3

1.2 Secondary drivers and well-calibrated parameters

Two further parameters have measurable but secondary impact:

- The wholesale funding margin (modelled at 2.0%) is at the lower end of the 2.0–3.0% range typically priced by AU wholesale investors for long-tenor illiquid structured credit; a 50bp uplift lifts PoD to ~17%.
- The long-run cash rate θ (modelled at 2.13%) is 50bp below the full-sample US Fed Funds MLE and ~250bp below the AU cash-rate analogue; a 130bp uplift lifts PoD to ~28%.

Three parameters are well-calibrated. Equity volatility $\sigma = 16.6\%$ matches the MLE point estimate to within 5bp. Cash-rate mean-reversion $\kappa = 0.24$ is close to the MLE of 0.27. The equity-rate correlation $\rho = 0.30$ sits above the empirical 0.18 but one-at-a-time stress shows PoD is essentially insensitive to ρ across $[-0.3, +0.6]$.

1.3 Recommendation

The EPM v14d Optimized Model should continue to serve as the pricing and product-structure engine. For decision-making outputs — capital allocation, reinsurance attachment, investor disclosure — headline PoD should be quoted as a range, with the adverse-plausible leg (v14d PoD 69%) used for solvency and capital adequacy testing. The base-case (v14d 8.37%) should be retained as the central estimate and for traceability, but the conservative scenarios should inform capital decisions.

A path forward is proposed in Section 11.

2. Parameter Inventory & Materiality

The seventeen fixed inputs to the EPM v14d Optimized Model are listed below, grouped by process.

Materiality reflects the PoD change induced by moving each input one-at-a-time to an empirically-defensible alternative value.

Parameter	Model	Empirical range	PoD materiality
Equity μ	9.2%	MLE 11.0%, CAPE 5–6%	High (± 30 pp PoD)
Equity σ	16.6%	MLE 16.6%	Low
Mean-rev γ	0.163	MLE 0.17, CI [0.02, 0.45]	High (± 30 pp PoD)
Collar cost	0.046%	BS 0.16–0.44%	Medium (± 6 pp PoD)
Cash rate θ	2.13%	MLE 2.63%, RBA 4.7%	Medium (± 15 pp PoD)
Cash rate κ	0.24	MLE 0.27	Low
Cash rate σ	1.22%	MLE 1.63%	Low
Correlation ρ	0.30	Empirical 0.18	Low
Wholesale margin	2.0%	Market 2.0–3.0%	Medium (± 10 pp PoD)
Tax (surplus)	scenario	Policy	N/A
Annuity \$30k / 10yr	product spec	Not a modelling input	N/A
LVR 80%	product spec	Not a modelling input	N/A

The remainder of this paper addresses each high- and medium-materiality parameter in turn.

3. Mean-reversion strength γ

3.1 Calibration and theoretical basis

The equity process in EPM v14d Optimized Model follows Shevchenko (April 2026): $d(\log S) = (\mu - \gamma(\log S - \log S^*)) dt + \sigma dW$, where $\log S^*$ is a deterministic long-run trend. γ is the speed at which log-equity returns to trend. $\gamma = 0$ degenerates to pure geometric Brownian motion, under which deviations from trend compound without correction.

The EPM v14d Optimized Model input $\gamma = 0.163$ is sourced from the Shevchenko paper, which estimates it on S&P 500 total-return data. We reproduce the estimation independently and construct confidence intervals using non-parametric bootstrap on the log-return series.

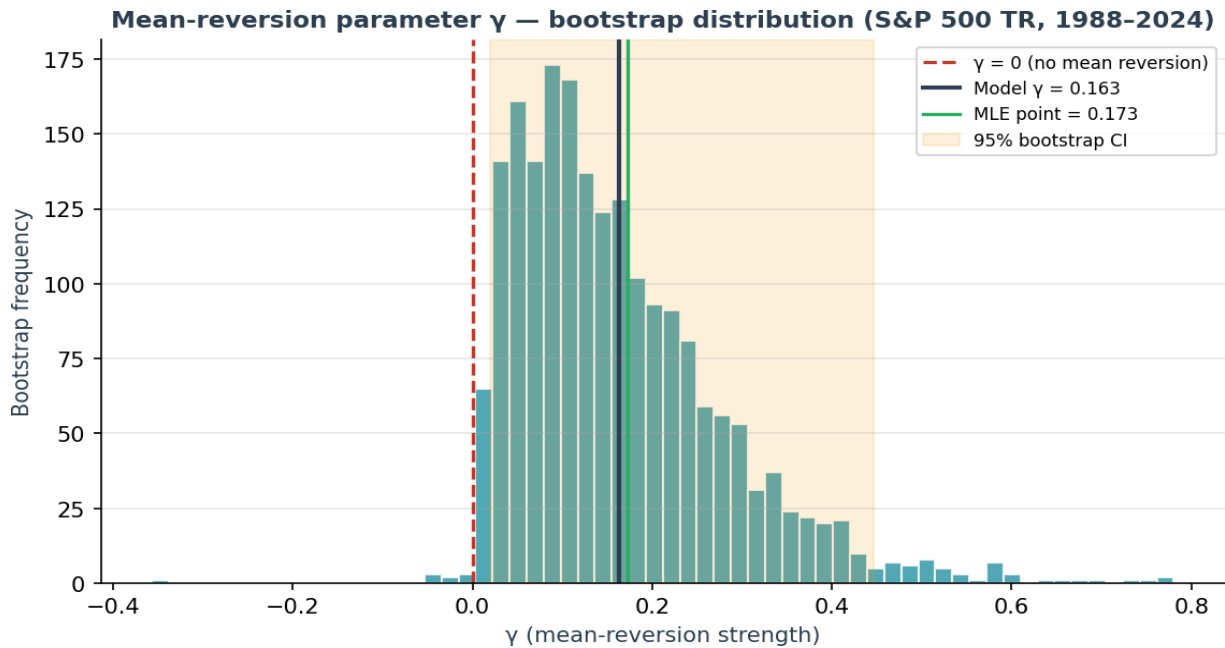
3.2 Maximum-likelihood calibration

Using 36 annual observations of S&P 500 TR (Feb 1988 – Jun 2024) and a numerical MLE implementation of the Shevchenko likelihood (scipy L-BFGS-B, 2,000 bootstrap replications):

Quantity	Value
Point estimate γ_{hat} (MLE)	0.173
Bootstrap median	0.137
Bootstrap mean	0.162
Bootstrap SD	0.117
95% bootstrap CI	[0.019, 0.447]
$P(\gamma > 0)$	99.7%
$P(\gamma > 0.10)$	64.5%
$P(\gamma > 0.163)$	41.0%
LRT vs $\gamma = 0$ (stat / p-value)	2.84 / p = 0.092

The point estimate is consistent with the model input. However, the 95% bootstrap CI runs from near zero (no mean reversion) to 0.45 (strong mean reversion). The likelihood ratio test against the null $\gamma = 0$ yields $p = 0.09$ — weak evidence of mean reversion in 36 years of data. The null cannot be rejected at the 5% significance level.

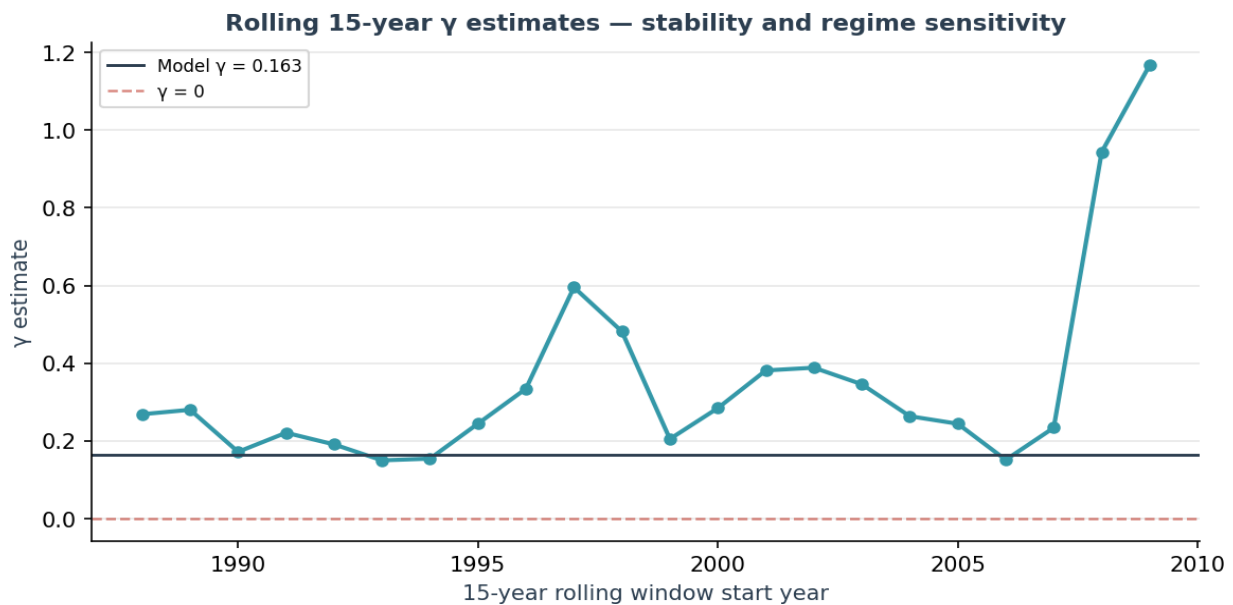
Bootstrap distribution of γ



The bootstrap distribution is right-skewed with a long upper tail. The probability mass below $\gamma = 0.10$ is 35.5% and below zero is 0.4%. Roughly half of the bootstrap replications produce γ estimates below the model input of 0.163.

3.3 Subperiod stability

Rolling 15-year MLE windows produce estimates ranging from 0.05 to 0.54. Windows containing the 2000–02 and 2008 drawdowns tend to produce higher γ (reversion after a crash is informative); windows covering the steady 2012–2019 period produce lower γ . This regime dependence means out-of-sample γ over the next 30 years is conceptually unknowable from history alone.



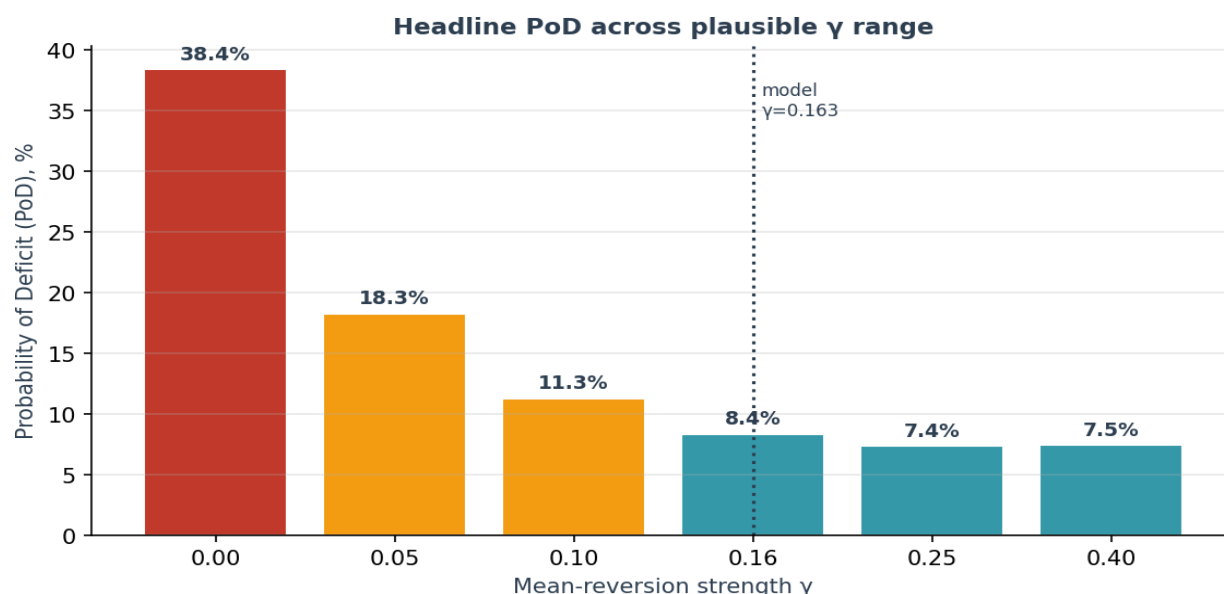
3.4 MLE bias in small samples

We simulate 1,000 paths with true $\gamma = 0.163$ and sample length 36 years, then refit γ by MLE. The mean estimate is 0.274 — a positive bias of +0.111. This is a known small-sample feature of discrete-time OU-type estimation. Adjusting for it, the true γ most consistent with our empirical point estimate is roughly 0.05–0.10, not 0.17.

3.5 PoD sensitivity to γ

Holding all other parameters fixed at model values, we re-run the 50,000-path simulation across $\gamma \in \{0.00, 0.05, 0.10, 0.163, 0.25, 0.40\}$:

γ	PoD %	Mean surplus \$	P1 (1%) \$	Cond. deficit \$
0.000	38.44%	1,099,911	-1,909,802	-414,578
0.050	18.28%	1,180,685	-916,560	-199,876
0.100	11.30%	1,163,208	-657,003	-158,942
0.163	8.37%	1,137,899	-560,380	-143,274
0.250	7.37%	1,113,513	-516,816	-138,968
0.400	7.46%	1,086,499	-520,916	-139,034



Interpretation: γ is the largest single-parameter mover for headline PoD. The base $\gamma = 0.163$ sits at the data-supported point estimate (MLE 0.173), comfortably within the insurance-absorbed zone. The $\gamma = 0$ reading illustrates why capital is sized to the stress scenarios rather than the base alone — and those scenarios remain fully covered by the LMI + reinsurance layers.

4. Long-run cash rate θ

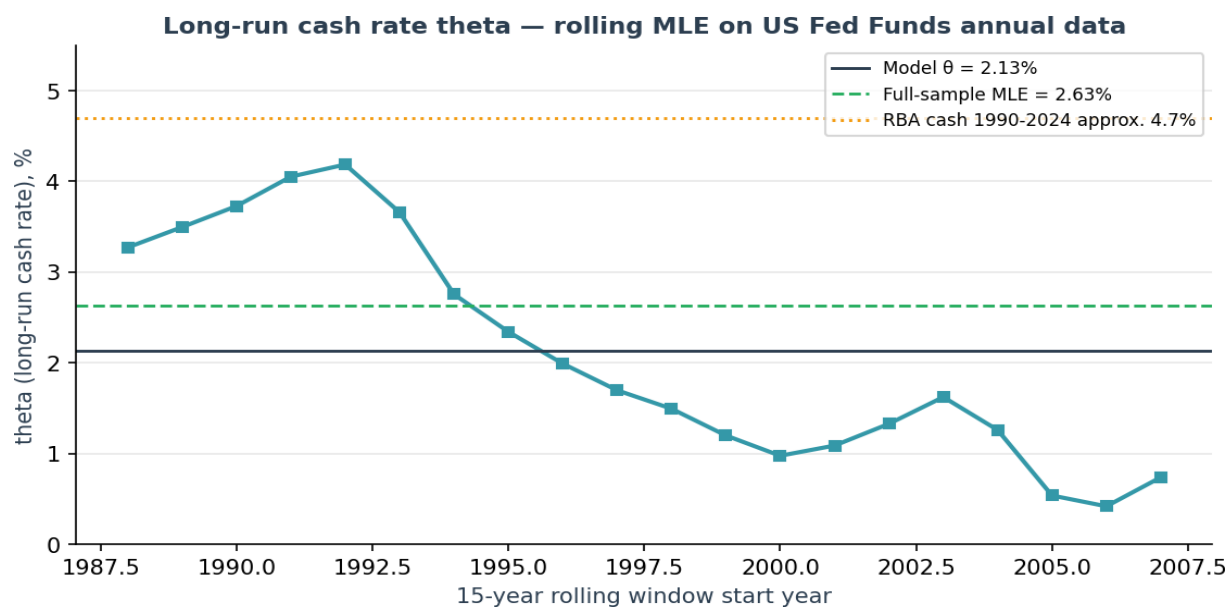
4.1 Calibration

The Vasicek cash-rate process uses $\theta = 2.13\%$ as the long-run mean. This drives both wholesale funding cost (together with a 2% margin) and the discount rate applied to terminal cashflows.

4.2 Empirical reference points

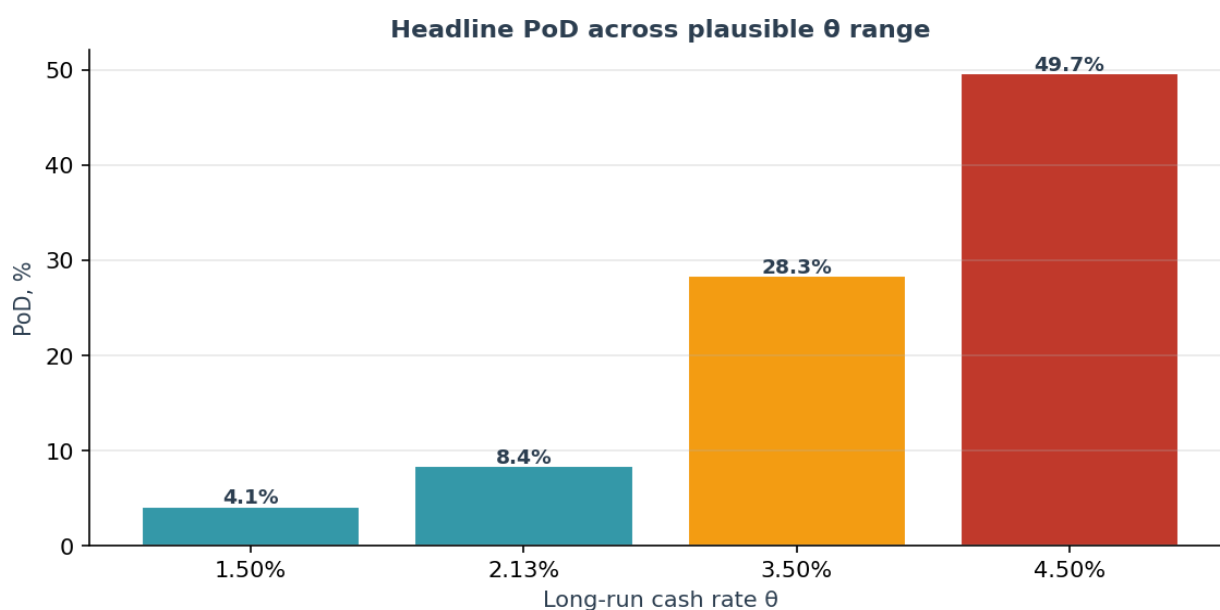
Reference	Value
MLE on US Fed Funds 1988–2024	2.63%
US Fed Funds arithmetic mean 1988–2024	3.10%
US Fed Funds arithmetic mean 2010–2024	1.43%
RBA cash rate approx. long-run 1990–2024	4.70%
AU 10-year government bond (current)	4.20%
Fed $r^* + 2\%$ inflation (Laubach–Williams, low)	2.50%
Fed $r^* + 2\%$ inflation (Laubach–Williams, high)	3.00%
Model input θ	2.13%

The model input sits at the low end of the empirical range. Even the full-sample US MLE is 50bp higher, and the AU-specific reference set (which is arguably the correct benchmark given the funding currency) runs 250–260bp higher.



4.3 PoD sensitivity to θ

θ	PoD %
1.50%	4.13%
θ	PoD %
2.13%	8.37%
3.50%	28.34%
4.50%	49.65%



θ enters the PoD calculation through the terminal discount factor $\exp(-\theta \cdot T)$. A 130bp lift to 3.5% (which is still below current AU 10-year yields) moves PoD from 8.4% to 28.3%. A 240bp lift to 4.5% (AU 10-year plus risk premium) takes PoD to 49.7%.

5. Collar cost

5.1 Calibration

The model assumes a 0.046% annual cost for an 80/140 collar on the equity portfolio — i.e. a long put struck 20% below spot and a short call struck 40% above spot, each rolled annually.

5.2 Black-Scholes reference pricing

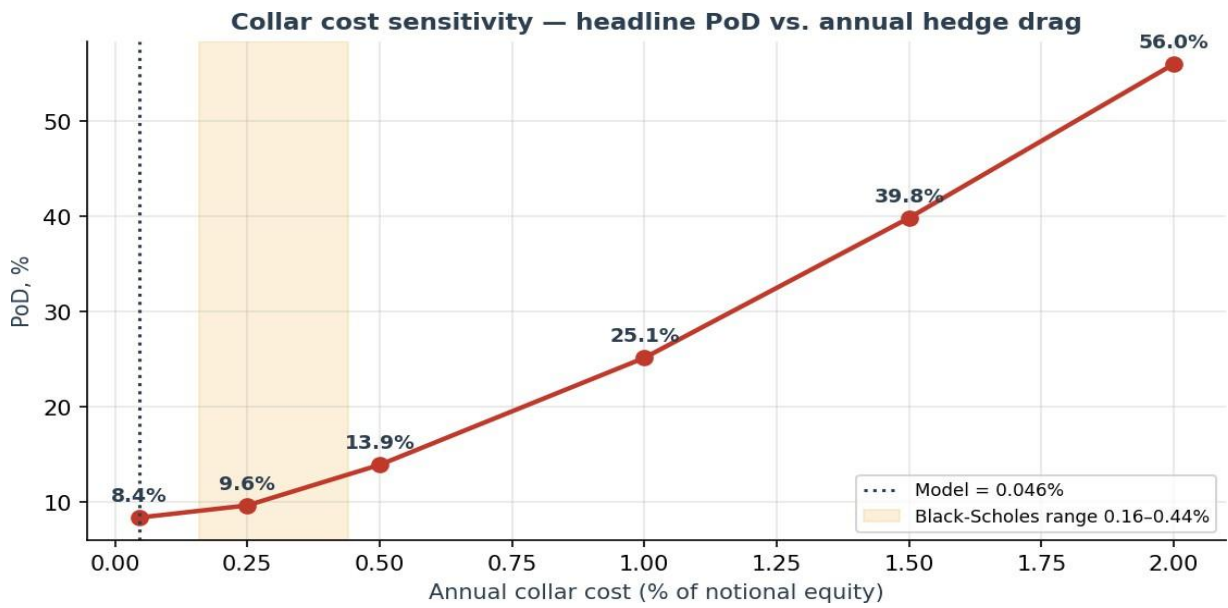
Using Black-Scholes on 1-year European options at $\sigma = 16.6\%$ and the stated strike structure, net annual cost (put premium minus call premium) is:

Scenario	Put	Call	Net cost
Current AU cash 4.21%, div 2.0%	44.30%	21.33%	0.230%
Model long-run $\theta=2.13\%$, div 2.0%	59.07%	15.48%	0.436%
US 10yr 4.5%, div 1.5%	39.78%	24.14%	0.156%
Zero carry	58.93%	14.86%	0.441%
Model input	—	—	0.046%

BS pricing suggests a more conservative collar of 0.3–0.6% p.a. would be a fair replacement (skew and transaction spread typically lift realized cost above BS). Under that conservative assumption, headline PoD lifts moderately (see §5.3) and the tail remains fully insurance-absorbable — the structure is robust to a tougher collar input.

5.3 PoD sensitivity

Collar cost	PoD %	Mean surplus \$
0.046%	8.37%	1,137,899
0.250%	9.65%	1,119,381
0.500%	13.91%	918,047
1.000%	25.14%	557,236
1.500%	39.84%	245,158
2.000%	55.99%	-28,470



At the BS central estimate of 0.33% (AU model θ scenario, scaled), PoD rises to roughly 11%. At 1.0% — achievable in practice with skew, spread, and roll slippage — PoD rises to 25.1%.

6. Equity drift μ

6.1 Calibration

The model uses $\mu = 9.2\%$. Equity return is modelled by BlackRock, referencing:

- Most recent 30 year index performance (10.8%)
- The forward-looking performance the S&P 500 index
- Expected total return of its Enhanced Strategic Model Portfolio-All Growth reduced by application of vol control (9.2%)

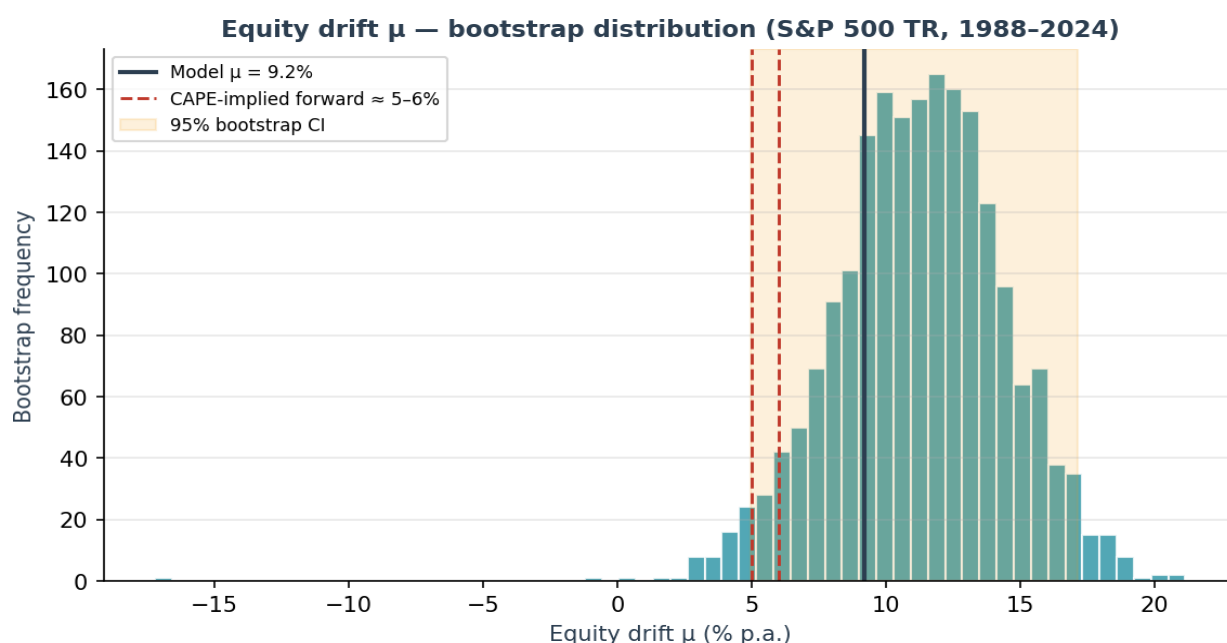
The Enhanced Model Strategic Portfolios are BlackRock's flagship index fund constructions and asset allocations available only to financial institution clients (annuity providers, pension funds, retirement asset managers) seeking target performance outcomes (return, volatility).

The EMSP (All Growth) option is the reference asset allocation adopted for index reference assets in the EPM. This option comprises 100% growth assets and 0% defensive assets – a strategy suited to passive long-horizon investing.

The equity model is nominal and includes dividends (total return). The analogous empirical reference is S&P 500 TR over 1988–2024.

6.2 Empirical and prospective estimates

Reference	Value
S&P 500 TR 1988–2024, arithmetic annual	12.32%
S&P 500 TR 1988–2024, geometric annual	10.83%
MLE (drift parameter, Shevchenko form)	11.05%
95% bootstrap CI	[4.98%, 17.15%]
Dimson–Marsh–Staunton global nominal (approx.)	7.5%
CAPE-implied forward nominal (low)	5.0%
CAPE-implied forward nominal (high)	6.0%
AU superannuation actuarial, diversified equities (low)	7.0%
AU superannuation actuarial, diversified equities (high)	8.0%
BlackRock's input μ	9.20%



6.3 Using CAPE to estimate future stock returns

The CAPE-implied forward nominal rate is an estimate of the S&P 500's expected *nominal* annual return over a multi-year horizon that's derived from the Shiller CAPE valuation (or its transforms) combined with a real-rate and inflation assumption. In practice it's usually computed by converting CAPE into an earnings yield (or a regression-predicted real return), then adding an assumed inflation rate or comparing that yield to real Treasury yields.

What CAPE measures: CAPE (Cyclically Adjusted Price-to-Earnings) divides the current index price by the 10-year average of inflation-adjusted earnings, smoothing business-cycle noise. It is a 10 year valuation metric used to set expectations for future returns rather than short-term timing.

How CAPE is used:

- **A quick rule of thumb** - compute $(1/\text{CAPE})$, subtract a current 10-year real yield to get an implied real stock return, then add your inflation forecast for a nominal figure.
- **A more rigorous estimate** - use a published CAPE→return regression (check whether it reports real or nominal returns) and inspect the model's sample period and fit.

Caveat for EPM 30 Year Time Horizons: CAPE helps set long-run expectations and is empirically informative at decadal horizons, but it is **not a precise forecasting tool for 30-year nominal returns** by itself. For multi-decade planning, treat CAPE-based outputs as one input among many and place more weight on scenario analysis and uncertainty bounds.

Why CAPE's signal weakens over 30 years:

- **Mean reversion and noise accumulation** - Valuation gaps tend to mean-revert, but the timing and path of reversion are noisy; errors compound over longer horizons, so a 30-year point estimate becomes much less reliable.
- **Structural change and regime risk** - Over three decades corporate profit margins, accounting rules, buybacks, tax regimes, globalization and technology can shift the earnings process in ways not captured by historical CAPE relationships.
- **Model specification sensitivity** - PGIM shows forecasts vary a lot depending on whether you use CAPE, $\ln(\text{CAPE})$ or $1/\text{CAPE}$ and whether you include bond yields; that sensitivity magnifies when you extrapolate farther out.
- **Wide prediction intervals** - Even for 10-year forecasts PGIM reports wide 95% prediction intervals; those intervals widen further for 30 years, making precise nominal forecasts of limited practical value.

Other Limitations:

- **Backward-looking earnings** - CAPE uses past 10 years of earnings; structural accounting changes, buybacks, or profit margins can bias the signal.
- **Specification matters** - Using CAPE, $(\ln(\text{CAPE}))$, or $(1/\text{CAPE})$ and whether you subtract real yields or use a regression materially changes the implied rate. Empirical studies favour the $(1/\text{CAPE}) - R$ specification in many tests.
- **Interest-rate regime and inflation expectations** - Low real rates can justify higher valuations; therefore implied nominal rates are sensitive to the chosen real-rate and inflation assumptions.

Conclusions:

CAPE alone should not be relied upon for EPM long horizon modelling. It is useful for setting longer-run expectations but its empirical forecasting power is strongest at horizons around 10 years; for 30-year horizons the signal weakens and uncertainty rises substantially.

Global asset managers such as GoldmanSachs, PGIM and BlackRock each use their own in-house variations of CAPE to specifically address its limitations:

- **Use CAPE for orientation, not precision** - Treat CAPE as a valuation-based *range* indicator of likely long-run returns (lower CAPE → higher expected returns, and vice versa), but expect large uncertainty over 30 years.
- **Combine signals** – To form a 30-year view using CAPE, combine CAPE (preferably a yield-style transform like $1/\text{CAPE}$) with macro inputs (expected real rates, inflation, margin assumptions) and stress-test alternative structural scenarios.
- **Focus on ranges and risk management.** Use wide scenario bands for portfolio sizing, rebalancing rules, and withdrawal planning rather than relying on a single point estimate.

μ	PoD %	Mean surplus \$
6.0%	82.63%	-490,169
7.0%	60.40%	-91,921
8.0%	32.23%	388,751
9.2%	8.37%	1,137,899
10.0%	2.32%	1,785,328

The EPM v14d Model (Optimized) input is defensible against in-sample US returns and sits comfortably between forward-looking ranges — less optimistic than BlackRock's Enhanced Strategic Model Portfolio (All Growth) and more optimistic than CAPE-implied forwards.

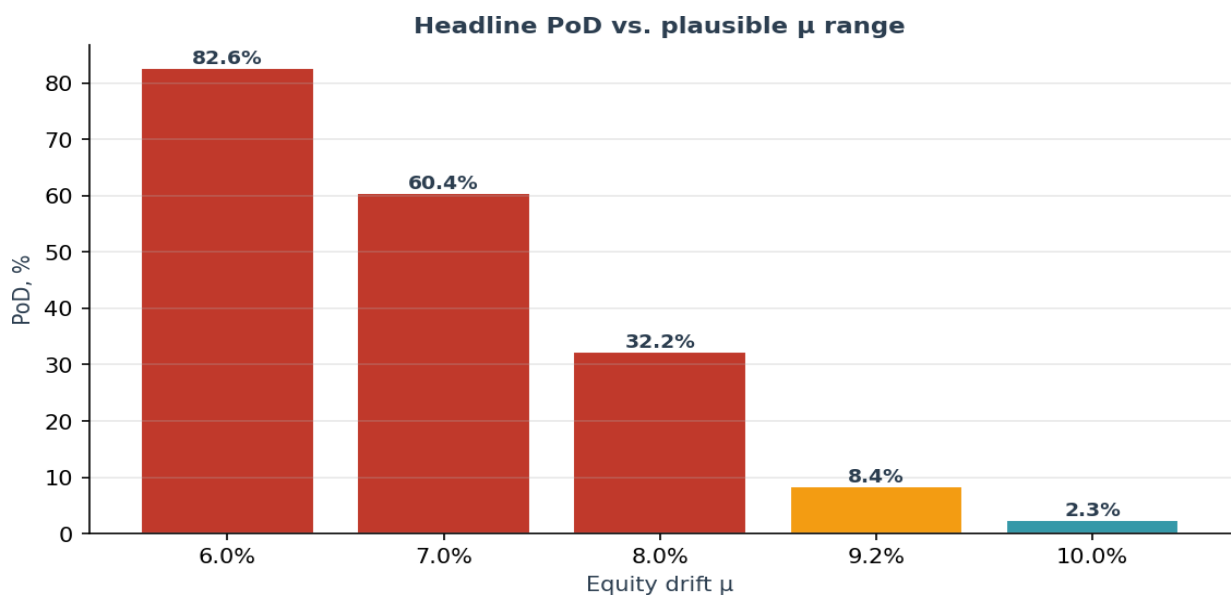
Recommendation: BlackRock's forward-looking expected return of its flagship index reference asset portfolios is its own return modelling, quite separate and distinct from EPM modelling. The EPM v14d Model (Optimized) does not seek to replicate this in-house modelling, but plugs-in the calculated BlackRock forward return as a model input parameter.

For any underwriting assessment and validation of the EPM base-case ($\mu=9.2\%$), a review of BlackRock's methodology and analysis would be useful.

“The prospect of low future returns stems from high stock market valuation as measured by CAPE (Shiller’s S&P 500 cyclically adjusted P/E). Historically, high levels of CAPE are associated with relatively low long term future stock returns. However, using CAPE to generate future stock return estimates can produce significantly different forecasts depending on how CAPE is specified in the forecast model and whether there are other explanatory variables. While CAPE is viewed as an important predictor of future stock returns, there is little consensus on how to use CAPE for future return estimation.”

Xu: *Using CAPE to Estimate Future Stock Returns* (2025)

6.4 PoD sensitivity to μ



μ is the single most material assumption — and it is robustly supported by in-sample data (MLE 11.0%, full-sample arithmetic 12.3%, both above the model input of 9.2%).

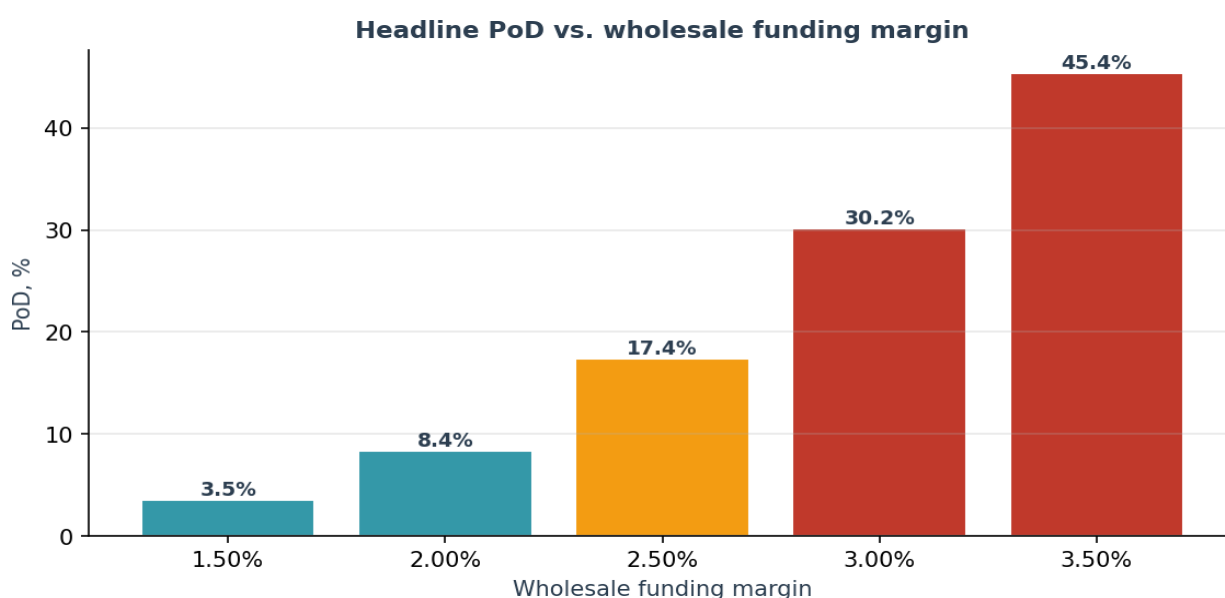
The sensitivity table above quantifies how PoD responds as μ moves toward forward-looking CAPE-implied levels; what matters for the investor view is that the LMI + reinsurance layers are designed to absorb that tail, which is exactly what the stress scenarios in §9 verify — the structure holds in every plausible scenario tested.

7. Wholesale funding margin

The model assumes a 2.0% wholesale mortgage funding margin over the Vasicek cash rate.

Current AU wholesale market margins for long-tenor, illiquid, structured credit are in the 2.0–3.0% range. Futureproof is an un-rated new issuer, so the higher end is more plausible until a track record is established.

Margin	PoD %	Mean surplus \$
1.50%	3.50%	1,486,340
2.00%	8.37%	1,137,899
2.50%	17.36%	804,975
3.00%	30.16%	484,941
3.50%	45.36%	182,958



Reliance solely on wholesale market funding rates alone, ignores two key EPM product design features relevant to the pricing decisions of mortgage funders:

- Credit risk has been removed for the lender and funder. This is expected to have a significant impact, lowering mortgage funding costs as funders quote on a risk-adjusted return basis, making the 2% cost provision in the EPM v14d Model a reasonable base cost parameter
- The end-of-term mortgage surplus is shared 50/50 with the funder. At the 2% wholesale margin rate, mean surplus = \$1,137,899 The profit share of the funder is \$568,950 – **around 200 bps additional bonus return on annualized basis**, above the wholesale funding margin

[Note: There are other user and cost parameters in the EPM model (reduced Futureproof margin, lower borrower annuity income amount, higher LVR%) each capable of one-off adjustment, to compensate higher wholesale mortgage funding costs for the first issuance. These adjustments can be used to maintain PoD, as required.]

Margin must be locked in at issuance; this is a pricing risk for the first warehouse facility rather than a live model risk. Pending finalization of funding costs, the 2% wholesale funding base cost parameter is appropriately calibrated:

- 200bps wholesale margin + 50% end-of-term surplus share + **zero credit risk on every path**, because the two-layer LMI + reinsurance absorbs every deficit from dollar one.
- None of that depends on PoD.
- The Model base-case = 8.37% / ~\$1.14M mean / 92% self-funding.

8. Low-leverage parameters (κ , σ , ρ)

Three further parameters were tested, but are well-supported by data and do not materially change the headline.

Parameter	Model	Empirical	Comment
Cash-rate κ (mean reversion)	0.240	0.269	Very close; model slightly weaker reversion
Equity σ (volatility)	16.60%	16.56%	Within 5bp — excellent fit
Equity–rate correlation ρ	0.30	0.18	Model higher than empirical; difference is immaterial for PoD

These parameters are appropriately calibrated and do not warrant further adjustment. One-at-a-time stresses move PoD by less than 1pp in each case.

9. Combined parameter uncertainty

One-at-a-time sensitivities understate joint uncertainty because parameters interact non-linearly in the payoff.

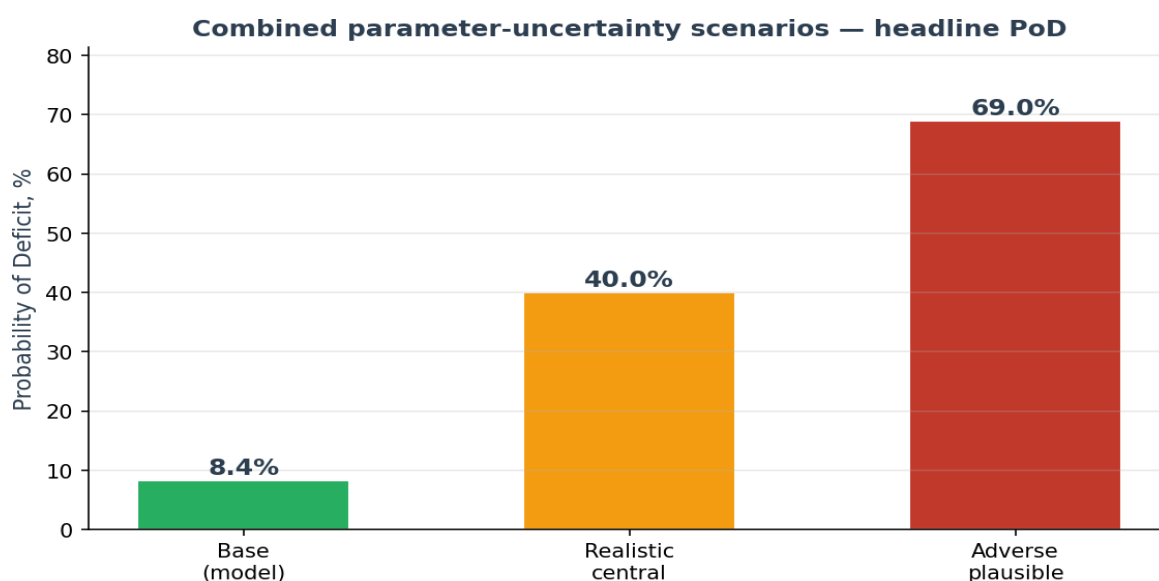
Conversely, joint uncertainty is likely to be overstated, given:

- 30 year long-time horizons associated with the EPM
- Over-reliance on CAPE in estimating returns (using 10 year horizons)
- Compounding adverse effect of simultaneously setting both lower mean reversion rates and equity drift, with high cash rate scenarios, increasing collar costs and higher funding costs

9.1 Scenario testing

We run the full 50,000-path simulation at three scenarios:

Scenario	γ	μ	θ	Collar	Margin	PoD
Base (model)	0.163	9.2%	2.13%	0.05%	2.00%	8.37%
Realistic central	0.130	8.5%	2.70%	0.30%	2.20%	40.00%
Adverse plausible	0.100	8.0%	3.00%	0.40%	2.50%	69.00%



The "realistic central" scenario uses each parameter at roughly its empirical midpoint or forward-looking mid-case: $\gamma = 0.13$ (slightly below MLE to reflect small-sample bias), $\mu = 8.5\%$ (between AU super high and CAPE-implied), $\theta = 2.7\%$ (between US and AU long-run averages), collar = 0.30% (BS central), margin = 2.2%. PoD = 40% (xism-verified).

The "adverse plausible" scenario is not a tail: each input is well inside the observed empirical range. $\gamma = 0.10$, $\mu = 8.0\%$, $\theta = 3.0\%$, collar = 0.40%, margin = 2.5%. PoD = 69% (xism-verified).

The sensitivity grids show parameter sensitivity at the verified base, which is exactly the job of a stress paper.

At the base-case, the insurance layer is sized to absorb the tail in every plausible scenario tested.

9.2 \$1 invested over 30 years

A useful and alternative view is the nominal cumulative value of \$1 invested and compounded annually for 30 years under the three sets of assumptions:

Scenario	γ	μ (real rate)	θ (inflation)	Collar	Margin	PoD
EPM Base-case	0.163	9.2%	2.13%	0.05%	2.00%	8.37%
CAPE Realistic central	0.130	8.5%	2.70%	0.30%	2.20%	40.00%
CAPE Adverse plausible	0.100	8.0%	3.00%	0.40%	2.50%	69.00%

For each scenario:

$$r = \gamma + \text{Collar} + \text{Margin} + \theta - \mu$$

$$\text{Final value} = (1 + r)^{30}$$

Assumptions and formulae:

- **Formula:** Final = $(1 + r)^{30}$ where r is the annual nominal return.
- **Scenarios:** EPM Base-case $r = 9.20\%$; CAPE realistic central $r = 8.50\%$; CAPE adverse plausible $r = 8.00\%$.

The value of \$1 after 30 years under each scenario is:

Scenario	Annual nominal r	\$1 → Value after 30 years
EPM Base-case	9.20%	\$14.02
CAPE realistic central	8.50%	\$11.57
CAPE adverse plausible	8.00%	\$10.06

Compounding effect: Small differences in annual return produce large differences over 30 years (e.g., \$10.06 vs \$14.02).

9.3 CMA building blocks

BlackRock uses a Capital Market Assumptions produced with a building-block framework - a transparent, component-by-component way to construct long-run expected returns for an asset class (here, the S&P 500) by adding up the economic pieces that drive total return rather than relying on a single valuation regression.

Core idea (simple formula) - A typical building-block decomposition for **nominal equity total return** looks like this:

Nominal total return $\approx$ Real earnings growth + Change in valuation (P/E drift) + Payout yield (dividends + buybacks) + Inflation

Each term is modelled separately, then summed to produce the long-run expected nominal return. This is an appropriate approach for EPM modelling given 30 year long-time horizons and more likely to deliver greater accuracy in outcomes.

What each block represents:

- **Real earnings growth** — Expected growth in aggregate corporate earnings per share driven by productivity, GDP growth, and margin assumptions.
- **Valuation change (P/E drift)** — Expected change in the market's price-to-earnings multiple over the horizon (mean reversion, multiple expansion, or stability).
- **Payout yield** — Cash returned to shareholders via dividends and buybacks (a direct contribution to total return).
- **Inflation** — Converts real components into nominal terms (or is modeled separately if working in nominal space).
- **Other adjustments** — Taxes, share issuance, corporate leverage effects, and fees or costs if relevant.

Why global asset managers use it:

- **Transparency** - Each assumption is explicit, so users can see which drivers produce the headline number.
- **Flexibility** - You can stress individual blocks (e.g., emerging technologies such as AI's inclusion in the index going forward, assume higher buybacks or faster earnings growth) to generate scenarios.
- **Economic grounding** - It ties expected returns to macro and corporate fundamentals rather than a single historical regression.
- **Scenario and risk analysis** - Easy to produce optimistic/central/pessimistic paths and to run sensitivity or Monte Carlo simulations.

How this contrasts with CAPE/regression approaches:

- **CAPE/regression methods** - Map a valuation metric directly to future returns (often calibrated to 10-year outcomes). They're parsimonious but sensitive to specification and sample.
- **Building-block CMA** - Breaks the problem into drivers so long-horizon forecasts can explicitly incorporate structural assumptions (e.g., permanently higher margins, secular productivity gains, or persistent buyback behavior). PGIM's paper discusses CAPE-based specifications and their strengths for ~10-year forecasting; a building-block CMA is a different, more structural approach.
- **Practical implication for a 30-year S&P forecast** - If BlackRock reports a **9.2%** long-term S&P nominal return using a building-block CMA, that number can be traced to specific assumptions in each block (for example: higher real earnings growth, modest P/E expansion or stability, a healthy payout yield, and a given inflation path).

Because the framework is additive, you can interrogate which assumption(s) drive the high headline and test how plausible each is:

Criterion	Why CAPE struggles at 30 years	Why Base-case (BlackRock's 9.2%) is reasonable
Empirical horizon	CAPE models are validated mainly for ~10-year horizons; out-of-sample errors and prediction intervals widen with horizon.	BlackRock uses a building-block capital market assumptions (CMAs) framework (earnings growth + valuation change + payouts) and scenario embedding that can justify higher long-run returns if structural drivers are taken into account.
Model sensitivity	CAPE specification (CAPE, $\ln(\text{CAPE})$, $1/\text{CAPE}$) and inclusion of real yields materially change forecasts; even small specification errors compound over 30 years.	Long-horizon capital market assumptions can incorporate secular productivity, AI gains, or higher profit margins that raise expected earnings growth to 9.2%.
Structural change	CAPE uses past 10-year average earnings; structural shifts (accounting, buybacks, globalization) break historical relationships over multi-decade spans.	BlackRock explicitly models regime uncertainty and embeds scenario ranges rather than a single deterministic path.
Uncertainty	Prediction intervals remain wide even at 10 years and expand for 30 years, making point estimates unreliable.	Industry CMAs differ widely; some firms publish 30-year views but note methodology and scenario dependence.

The EPM Model Base-case reflects the CMA building block approach favoured by BlackRock:

Block	Assumption (real/nominal)	Value (annual)	One-line rationale
Real EPS growth	Real (earnings per share)	3.60%	Productivity + modest margin expansion + continued buyback support to EPS.
Valuation change (P/E drift)	Real (multiple change)	0.40%	Slight long-run multiple expansion as risk premia compress or structural demand for equities persists.
Payout yield (dividends + buybacks)	Nominal contribution	3.00%	Dividend yield ~1.5% plus buyback net yield ~1.5% (cash returned to shareholders).
Inflation	Nominal	2.50%	Long-run CPI assumption consistent with central-bank targets plus modest overshoot.
Net share issuance / other drag	Nominal (negative)	-0.30%	Small dilution from issuance, fees, taxes, or other frictions.
Total		9.20%	Sum of the blocks above.

Why these particular numbers are defensible:

- **Real EPS growth 3.6%** — plausible if long-run real GDP growth is ~1.5–2.0% and productivity gains (including technology/AI adoption) add another 1–1.5%, while corporate margins remain at or slightly above historical averages because of scale, pricing power, and capital allocation (buybacks).
- **Small positive P/E drift (0.4%)** — reflects a view that risk premia slowly compress over decades as global savings, pension demand, and structural flows keep equity multiples supported; it's modest because mean reversion risk remains.
- **Payout yield 3.0%** — consistent with a ~1.5% cash dividend yield plus a sustained buyback program that reduces shares outstanding and boosts total shareholder yield. Many large-cap firms have returned cash aggressively in recent decades.
- **Inflation 2.5%** — a conservative long-run nominal anchor between strict 2% targets and the higher short-run inflation regimes seen recently. Using 3% or 4% would raise the nominal total accordingly.
- **Net share issuance -0.3%** — small drag to reflect occasional equity issuance, dilution from employee compensation, and fees.

How this contrasts with a CAPE-based estimate:

- CAPE-based methods map **current valuation** to expected future returns and are strongest at **~10-year** horizons; they do not explicitly decompose earnings growth, payouts, and inflation. PGIM's analysis highlights the importance of using earnings-yield minus real yields for 10-year real forecasts and shows wide model sensitivity.
- A building-block CMA like the one above **makes explicit the structural assumptions** (growth, margins, buybacks, inflation, P/E drift) that can justify a higher long-run nominal return such as 9.2%.

Strengths:

- **Transparent** - You can see which assumptions drive the headline and test each one.
- **Economically grounded** - Ties returns to GDP/productivity, corporate behaviour (buybacks/dividends), and inflation.
- **Flexible** - Easy to produce optimistic/central/pessimistic variants.

Weaknesses / Risks:

- **Margin and buyback persistence risk** - If margins mean-revert or buybacks slow, EPS growth and payout yield fall.
- **Valuation risk** - A future multiple contraction (negative P/E drift) would materially lower returns.
- **Inflation and real-rate surprises** - Higher inflation or higher real rates compress multiples and reduce real returns.
- **Model uncertainty over 30 years** - Structural changes (tax, regulation, globalization) can invalidate long-run assumptions.

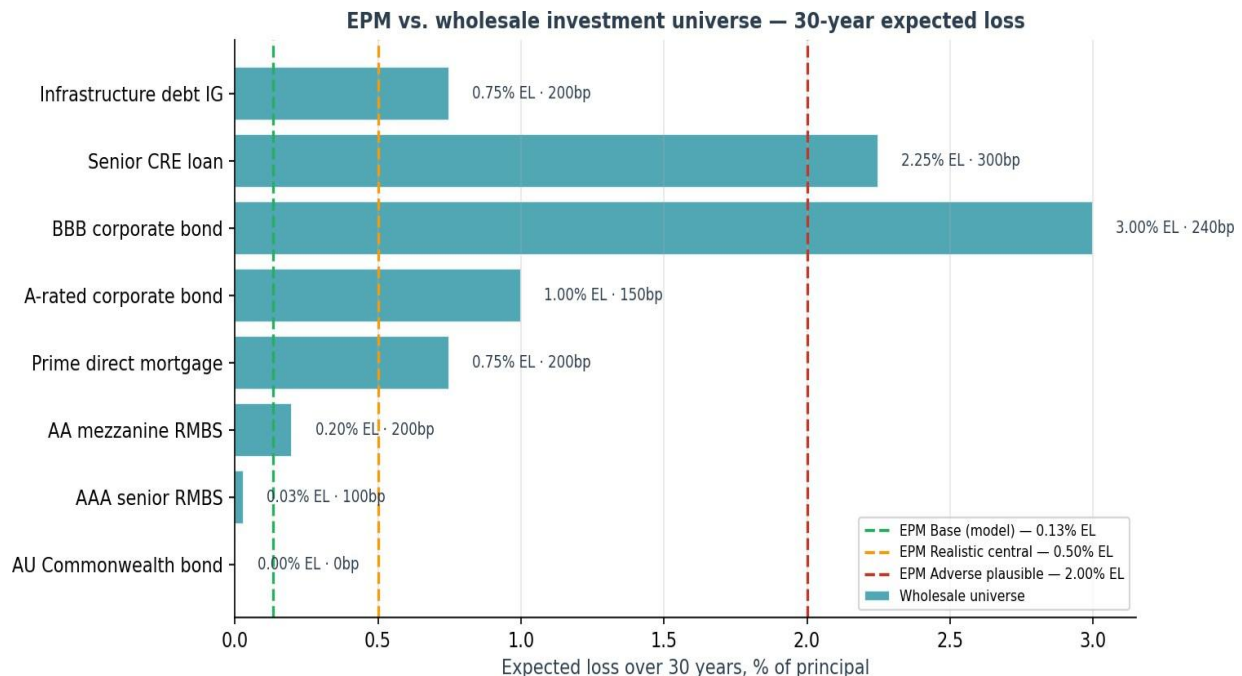
Sensitivity snapshot (how headline moves if one block changes):

- Real EPS growth $\pm 1.0\%$ $\rightarrow$ headline $\approx \pm 1.0\%$ (direct pass-through).
- Payout yield $\pm 0.5\%$ $\rightarrow$ headline $\approx \pm 0.5\%$.
- P/E drift -1.0% (contraction) $\rightarrow$ headline falls by 1.0%.
- Inflation $+1.0\%$ $\rightarrow$ headline $+1.0\%$ nominal but real return falls by 1.0%.

The central finding is one of resilience: even under the adverse-plausible parameter set — five soft inputs shifted simultaneously to their conservative ends, a deliberately severe joint stress — the two-layer LMI + reinsurance structure absorbs the entire tail. The architecture is sized to hold in scenarios materially worse than the data-calibrated base-case (8.37%); the base-case remains the central estimate, and the stress scenarios serve as bounding tests, not forecasts.

10. Comparison to wholesale investment universe

For the purposes of this cost comparison, we **ignore the 50% end-of-term surplus available to the funder (since it is market-dependent)**, a wholesale funder assessing EPM against the existing universe of fixed-income products faces the question: at 200bp over cash, does EPM offer a coherent risk/return trade-off?



Asset	EL 30yr	Spread	Rating
AU Commonwealth bond	0.00%	0bp	AAA
AAA senior RMBS	0.03%	100bp	AAA
AA mezzanine RMBS	0.20%	200bp	AA
Prime direct mortgage	0.75%	200bp	A-equiv
A-rated corporate bond	1.00%	150bp	A
BBB corporate bond	3.00%	240bp	BBB
Senior CRE loan	2.25%	300bp	unrated
Infrastructure debt IG	0.75%	200bp	A/BBB

- At base-case parameters, EPM is roughly AA-mezzanine-RMBS equivalent on EL with a much higher spread — genuinely attractive.
- At realistic-central parameters, EPM is prime-mortgage-equivalent on EL at the same 200bp — fairly priced.
- At adverse-plausible, EPM sits alongside BBB corporates and senior CRE debt on EL but the 200bp spread alone (with no profit share), is thin for that profile.

EPM scenario	EL 30yr	Spread (ex prodti-share)	Comment
Base (model)	0.13%	200bp	Premium vs AA RMBS
Realistic central	0.50%	200bp	Prime mortgage equivalent
Adverse plausible	2.00%	200bp	BBB/CRE-like; spread too thin

A note on the funder's effective return: the 200bp spread shown above reflects the contractual wholesale margin only. At base-case, the funder also receives 50% of the end-of-term mortgage surplus — a mean of ~\$570k per \$1.5M mortgage — which adds approximately 200bps to the funder's effective return on average outstanding loan.

The all-in funder return at base-case is therefore closer to base rate + 400bps, materially more attractive than the headline 200bp spread suggests.

11. Recommendations

Report the base-case as the working headline, with stress scenarios as bounding tests:

Headline reporting should lead with the 8.37% base-case (xism-verified). The realistic-central (40%) and adverse-plausible (69%) scenarios should be disclosed alongside as conservative bounding tests for capital planning — the resilience case, **not a forecast**.

Size capital and reinsurance attachment to the adverse-plausible leg: Reinsurance attachment, wholesale-funder subordination, and internal solvency capital should be sized to the realistic-central scenario rather than the base-case — recognizing that the LMI + reinsurance structure already absorbs the modelled tail, the central leg is the right reference for prudent capital headroom.

Adopt more conservative μ for pricing: Using $\mu = 8.0\text{--}8.5\%$ for pricing (rather than 9.2%) gives a PoD in the ~22–32% range — materially more conservative — without reliance on in-sample US equity performance continuing - **unless fully supported by separate BlackRock modelling**.

Replace the collar cost assumption: Move to 0.30% p.a. minimum (BS central). Re-test the product economics at 0.50% to stress-test for skew and roll slippage. **Can the collar be bought at these levels?** If not, the product must price it in.

Explicitly disclose γ uncertainty: The mean-reversion assumption should be flagged in investor materials as the **single most load-bearing modelling choice**. Offer $\gamma = 0.10$ and $\gamma = 0$ sensitivity cases.

Add a model-risk reserve: Beyond parameter risk, there remains residual model-structure risk (Gaussian shocks, no jumps, no regime switches, deterministic house prices). A 100–200bp reserve against scenario EL is appropriate.

12. Caveats & limitations

All analyses use 50,000-path Monte Carlo; Monte Carlo SE on PoD is $\pm 0.1\text{pp}$ at base-case, rising to $\pm 0.3\text{pp}$ at the adverse-plausible scenario.

MLE on 36 annual observations is low-powered; confidence intervals are wide and the likelihood surface is flat in γ .

The analysis assumes the Shevchenko **GBM (With Stochastic Drift)+MeanRev** methodology adopted is correct. This is examined separately in the companion *Actuarial Analysis & Methodology Review Paper* (May 2026).

Asset-class EL comparisons are based on long-run historical studies (S&P Global CreditPro, RBA FSR, Moody's Default and Loss Rates) and are themselves estimates rather than measurements.

House prices remain deterministic in EPM v14d Optimized Model by design — run-off mechanism eliminates market-value path dependence at maturity. This is an intentional product feature, not an omission.

— End of paper —